1. Clinical Question & Core Hypothesis

Clinical Question

Pretreatment TLS presence and organization should predispose MIBC to benefit from intravesical BCG, but not all TLS-rich tumors respond — what molecular or spatial constraints determine when TLS amplify BCG-elicited anti-tumor immunity? 2) Tier 1 paths used: I built on the Ontologist's core paths that TLS presence/density PREDICTS BCG_response, TLS proximity-to-tumor modulates immune impact, and spatial organization metrics reflect effective local immunity. These provide the mechanistic backbone linking morphology/spatial context to therapeutic amplification by BCG. 3) Tier 2 evidence supporting the claim: chemokine drivers of TLS (CXCL13, CCL21) and markers of B-cell zones (CD20, CD21) were listed as supporting entities; these explain how TLS form and organize to position effector and antigen-presenting cells near tumor epithelium. I therefore selected chemokines as mechanistic anchors because they directly mediate TLS recruitment/maturation and can explain TLS functionality. 4) Why therapy_response endpoint: BCG response (recurrence-free outcome or categorical response) is the most clinically relevant and proximal endpoint for this biomarker concept — the stated goal is to predict and explain differential BCG efficacy, not long-term OS. 5) Deliberate exclusions: I excluded broad spatial metrics and multiple immune-subtype markers from the core hypothesis to keep it mechanistically sharp (max 1 imaging feature + up to 2 anchor genes). Metrics like Ripley's K, tumor-budding, and additional IHC markers (CD3, CD8) are retained as secondary_modifiers/assays for validation but not in the core mechanistic claim?

Working Hypothesis

Pretreatment TLS-like lymphoid aggregates located proximal to the tumor boundary form a chemokine-rich niche dominated by CXCL13 and CCL21 that supports local antigen presentation and rapid recruitment/interaction of B cells (CD20+/CD21+ zones) and helper T cells, thereby amplifying the intravesical BCG–driven anti-tumor immune response. When TLS are present and CXCL13/CCL21 signaling is high, patients will exhibit superior therapy_response to intravesical BCG, with lower recurrence rates and longer recurrence-free survival; when TLS are sparse, poorly organized, or CXCL13/CCL21 signaling is low, BCG-induced immunity is insufficient, leading to poorer therapy_response. Specific, testable predictions include: (1) pretreatment TLS density and proximity to tumor correlate positively with favorable therapy_response; (2) pretreatment expression levels of CXCL13 and CCL21 correlate with TLS maturation/organization and predict therapy_response beyond TLS density alone; (3) ex vivo or in vitro disruption of CXCL13–CXCR5 or CCL21–CCR7 signaling diminishes BCG-stimulated immune activation in bladder tissue models; (4) CD20/CD21 prevalence within TLS serves as a surrogate for TLS functionality and strengthens the predictive value of CXCL13/CCL21. Experimental approaches will include retrospective tissue analyses (IHC for TLS markers, CXCL13, CCL21; CD20/CD21; CD3) with digital spatial quantification, RNA in situ hybridization or spatial transcriptomics for chemokine/cell-type localization, and correlation with therapy_response (responder vs non-responder) and recurrence-free survival; prospective validation could utilize similar assays in a new BCG-treated cohort, supplemented by ex vivo bladder tissue cultures to test chemokine-blockade effects on BCG-driven cytokine production and cell recruitment.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	CXCL13, CCL21
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CXCL13: CXCL13, CCL21) and markers of B-cell zones (CD20, CD21) were listed as supporting entities; these explain how TLS form and organize to position effector and antigen-presenting cells near tumor epithelium.

CCL21: CCL21) and markers of B-cell zones (CD20, CD21) were listed as supporting entities; these explain how TLS form and organize to position effector and antigen-presenting cells near tumor epithelium.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	MS4A1 (CD20)	Supporting modifier from knowledge graph
Secondary	CR2 (CD21)	Supporting modifier from knowledge graph
Secondary	CD8A	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: provide validation of TLS quality without diluting the core mechanistic claim.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 3: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MS4A1 (CD20): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- CR2 (CD21): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.